SI Session Planning Form

SI Leader: Brandon Cobb
Session Week: 4
Session#: 4
Instructor: Professor Tramontozzi
Old Plans: SharePoint
Session Goal: Students will have reviewed unit conversions, algebraic manipulation of exact and inexact numbers, calculating their significant figures after multi-step operations like addition and multiplication. Student will review how to write scienticific notation. Students are given a reference sheet for conversion factors. Students will become adept with atomic particles and subatomic structures like orbitals of electrons.

Opener

Take Attendance

Collaborative Learning Techniques	Clusters
Learning Strategy	Send a Problem
Time	12:00—12:10
Materials	A writing utensil and index cards.
Lecture/Textbook/Old Plans/Slide References	Class roster, Chapter 2: Measurments, Chapter 3: Periodic Table

Definitions

1. **Clusters:** In Clusters, group participants are divided into smaller groups for discussion. They may also be allowed to self-select the small group they want to be in. After discussing the assigned topic, the cluster may report their findings to the large group.

2. **Send a Problem:** Generate a list of problems. Assign a different problem to each student. Give the students a minute to complete the first step of their assigned problem. After the minute, students pass their problem to the students to their right, who then complete the second step of the problem. Continue passing the problem until the steps are completed. After, have the students present their originally-assigned problems with the completed solutions to the entire group.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

Send a Problem (Orbitals Ice Breaker): Generate a list of chemistry problems of varying difficulty. Assign each student a different problem to solve. Give students one minute to complete the first step of their assigned problem. After the minute, each student passes their problem to the group on their right. The receiving group then completes the next step of the problem. This continues until all steps are completed. Finally, each student presents their originally-assigned problem along with the completed solution to the entire class.

1. The SI leader pairs up students based on proximity seating. If a student does not have a partner, the leader joins them.
2. Each pair is assigned a small chemistry task (e.g., determining concentration, calculating molar mass, predicting reaction products) suitable for a first step in a multi-step problem.
3. Pairs have two minutes to decide on their approach and collect any necessary tools or reference materials provided by the leader.
4. Pairs complete the first step of their problem, recording results clearly, and then pass the problem and any physical materials (if applicable) to the next group.
5. The next group evaluates the previous step and completes the subsequent step. If a group identifies an incorrect approach or calculation, they are allowed to correct it for the next step.
6. After all steps are completed, each original problem owner presents the full, completed solution to the class, highlighting any corrections or insights from peer contributions.
7. The SI leader facilitates discussion on problem-solving strategies, peer collaboration, and common chemistry misconceptions.

Students engage collaboratively, seeing multiple approaches to the same problem, learning from their peers, and practicing stepwise problem-solving on a shared "centralized canvas" of the problem.

Content (fully worked out problems & answers) continued...

(1) Calculate the atomic mass of carbon given its common isotopes: ^{12}C has mass 12.000000 amu (98.930%) and ^{13}C has mass 13.003355 amu (1.070%).

a. 12.001 amu

b. 12.01 amu

c. 12.10 amu

d. 13.003 amu

Solution:
Convert abundances to decimals: 0.9893 and 0.0107.

$$\text{atomic mass} = (12.000000)(0.9893) + (13.003355)(0.0107)$$

Compute each term:

$$(12.000000)(0.9893) = 11.87$$

$$(13.003355)(0.0107) = 0.1391$$

$$\text{atomic mass} = 11.87 + 0.1391 = 12.01 \text{ amu}$$

Answer: b. 12.01 amu

Content (fully worked out problems & answers) continued...

(2) Calculate the atomic mass of chlorine given its common isotopes: ^{35}Cl mass = 34.96885 amu (75.78%), ^{37}Cl mass = 36.96590 amu (24.22%).

a. 35.00 amu

b. 35.178 amu

c. 35.45 amu

d. 36.000 amu

Solution:
Convert abundances: 0.7578 and 0.2422.

$$\text{atomic mass} = (34.96885)(0.7578) + (36.96590)(0.2422)$$

Compute each term:

$$(34.96885)(0.7578) = 26.50$$

$$(36.96590)(0.2422) = 8.950$$

$$\text{atomic mass} = 26.50 + 8.950 = 35.45 \text{ amu}$$

Answer: c. 35.45 amu

Content (fully worked out problems & answers) continued...

(3) Calculate the atomic mass of magnesium using three common isotopes: ^{24}Mg mass = 23.985042 amu (78.99%), ^{25}Mg mass = 24.985837 amu (10.00%), ^{26}Mg mass = 25.982593 amu (11.01%).

a. 24.31 amu

b. 24.000 amu

c. 25.000 amu

d. 23.985 amu

Solution:
Convert abundances: 0.7899, 0.1000, 0.1101.

$$\text{atomic mass} = (23.985042)(0.7899) + (24.985837)(0.1000) + (25.982593)(0.1101)$$

Compute each term:

$$(23.985042)(0.7899) = 18.95$$

$$(24.985837)(0.1000) = 2.499$$

$$(25.982593)(0.1101) = 2.861$$

$$\text{atomic mass} = 18.95 + 2.499 + 2.861 = 24.31 \text{ amu}$$

Answer: a. 24.31 amu

Content (fully worked out problems & answers) continued...

(4) Calculate the atomic mass of silver from its two stable isotopes: ^{107}Ag mass = 106.90509 amu (51.839%), ^{109}Ag mass = 108.90476 amu (48.161%).

a. 107.000 amu

b. 107.868 amu

c. 108.000 amu

d. 106.905 amu

Solution:
Convert abundances: 0.51839 and 0.48161.

$$\text{atomic mass} = (106.90509)(0.51839) + (108.90476)(0.48161)$$

Compute each term:

$$(106.90509)(0.51839) = 55.427$$

$$(108.90476)(0.48161) = 52.441$$

$$\text{atomic mass} = 55.427 + 52.441 = 107.868 \text{ amu}$$

Answer: b. 107.868 amu

Content (fully worked out problems & answers) continued...

(5) Calculate the atomic mass of bromine given: ^{79}Br mass = 78.918337 amu (50.690%), ^{81}Br mass = 80.916291 amu (49.310%).

a. 79.000 amu

b. 80.500 amu

c. 79.903 amu

d. 78.918 amu

Solution:
Convert abundances: 0.50690 and 0.49310.

$$\text{atomic mass} = (78.918337)(0.50690) + (80.916291)(0.49310)$$

Compute each term:

$$(78.918337)(0.50690) = 39.999$$

$$(80.916291)(0.49310) = 39.904$$

$$\text{atomic mass} = 39.999 + 39.904 = 79.903 \text{ amu}$$

Answer: c. 79.903 amu

Content (fully worked out problems & answers) continued...

(6) Which group of the periodic table do the alkali metals belong to?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
Alkali metals include Li, Na, K, Rb, Cs, and Fr. These elements are found in the first vertical column of the periodic table, which is Group 1.

Answer: a. Group 1

Content (fully worked out problems & answers) continued...

(7) Calcium (Ca) is classified as which type of element?

a. Alkali metal

b. Alkaline earth metal

c. Halogen

d. Noble gas

Solution:
Calcium (atomic number 20) is found in Group 2 of the periodic table. Group 2 elements are called alkaline earth metals.

Answer: b. Alkaline earth metal

Content (fully worked out problems & answers) continued...

(8) Which period does chlorine (Cl) belong to?

a. Period 2

b. Period 3

c. Period 4

d. Period 5

Solution:
Chlorine (atomic number 17) has its outer electrons in the 3rd energy level, so it belongs to Period 3.

Answer: b. Period 3

(9) Which of the following is a noble gas?

a. Oxygen (O)

b. Fluorine (F)

c. Neon (Ne)

d. Sodium (Na)

Solution:
Noble gases are found in Group 18. Neon (Ne) is in Group 18, while oxygen and fluorine are nonmetals in Groups 16 and 17, and sodium is an alkali metal in Group 1.

Answer: c. Neon (Ne)

Content (fully worked out problems & answers) continued...

(10) Which element is a metalloid?

a. Boron (B)

b. Magnesium (Mg)

c. Argon (Ar)

d. Bromine (Br)

Solution:
Metalloids have properties of both metals and nonmetals. Boron (B) lies along the staircase line and is classified as a metalloid.

Answer: a. Boron (B)

Content (fully worked out problems & answers) continued...

(11) Which group contains the halogens?

a. Group 1

b. Group 2

c. Group 17

d. Group 18

Solution:
The halogens include fluorine, chlorine, bromine, iodine, and astatine. These are located in Group 17 of the periodic table.

Answer: c. Group 17

Content (fully worked out problems & answers) continued...

(12) Which of the following is a transition metal?

a. Iron (Fe)

b. Sodium (Na)

c. Sulfur (S)

d. Neon (Ne)

Solution:
Transition metals are found in Groups 3–12 of the periodic table. Iron (Fe) is in Group 8, making it a transition metal.

Answer: a. Iron (Fe)

Content (fully worked out problems & answers) continued...

(13) Which element is in Group 2, Period 4?

a. Calcium (Ca)

b. Magnesium (Mg)

c. Beryllium (Be)

d. Strontium (Sr)

Solution:
Group 2 elements are alkaline earth metals. In Period 4, the Group 2 element is Calcium (Ca).

Answer: a. Calcium (Ca)

Content (fully worked out problems & answers) continued...

(14) Which of the following elements is a nonmetal?

a. Aluminum (Al)

b. Phosphorus (P)

c. Iron (Fe)

d. Calcium (Ca)

Solution:
Phosphorus (P) is a nonmetal located in Group 15. The others are metals: aluminum (metal), iron (transition metal), and calcium (alkaline earth metal).

Answer: b. Phosphorus (P)

Content (fully worked out problems & answers) continued...

(15) Which element is in Period 2, Group 18?

a. Helium (He)

b. Neon (Ne)

c. Argon (Ar)

d. Krypton (Kr)

Solution:
Period 2 corresponds to the second row of the periodic table. In Group 18 of this row, the element is Neon (Ne).

Answer: b. Neon (Ne)

Content (fully worked out problems & answers) continued...

(16) Write both the longhand and shorthand (noble-gas) electron configurations for neutral chromium (Cr).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

Solution:
Chromium, Cr, has atomic number $Z = 24$ so there are 24 electrons in the neutral atom. Fill orbitals in order (Aufbau) but note the well-known exception for Cr: a half-filled d subshell+a single s electron is more stable than $4s^2 3d^4$. Thus electrons arrange as:
Longhand (write from lowest to highest energy):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$$

Check electron count: $2 + 2 + 6 + 2 + 6 + 1 + 5 = 24$.
Shorthand: the core through 3p is argon: [Ar]. So shorthand is: [Ar]$4s^1 3d^5$.

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; [Ar]$4s^1 3d^5$

Content (fully worked out problems & answers) continued...

(17) Write both the longhand and shorthand electron configurations for Fe^{3+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$; Shorthand: [Ar]$4s^1 3d^5$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$; Shorthand: [Ar]$4s^2 3d^4$

Solution:
Neutral iron, Fe, has $Z = 26$ (26 electrons). Fe^{3+} has lost 3 electrons, so electrons remaining $= 26 - 3 = 23$. When removing electrons from transition metals remove $4s$ electrons before $3d$. Start from neutral Fe configuration: neutral Fe often written [Ar]$4s^2 3d^6$. Remove two $4s$ electrons first (-¿ $4s^0 3d^6$), then remove one $3d$ electron (-¿ $3d^5$). So the result is:
Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ Check count: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: c. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(18) Write both the longhand and shorthand electron configurations for sulfide ion, S^{2-}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^4$; Shorthand: [Ne]$3s^2 3p^4$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6$; Shorthand: [Ar]

c. Longhand: $1s^2 2s^2 2p^6 3s^1 3p^6$; Shorthand: [Ne]$3s^1 3p^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^5$; Shorthand: [Ne]$3s^2 3p^5$

Solution:
Neutral sulfur S has $Z = 16$ (16 electrons). S^{2-} gains 2 electrons $\Rightarrow$ total electrons $= 16 + 2 = 18$. The configuration for 18 electrons is the same as argon. Longhand filling:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$$

Check count: $2 + 2 + 6 + 2 + 6 = 18$.
Shorthand: the core is argon, so [Ar]. (One can also write [Ne]$3s^2 3p^6$ but [Ar] is simplest.)

Answer: b. $1s^2 2s^2 2p^6 3s^2 3p^6$; [Ar]

Content (fully worked out problems & answers) continued...

(19) Write both the longhand and shorthand electron configurations for neutral copper (Cu).

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; Shorthand: [Ar]$3d^{10} 4s^1$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$; Shorthand: [Ar]$4s^2 3d^9$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^0 3d^{11}$; Shorthand: [Ar]$3d^{11}$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^9$; Shorthand: [Ar]$4s^1 3d^9$

Solution:
Copper, Cu, $Z = 29$ so neutral Cu has 29 electrons. The observed (and most stable) configuration is an exception to the naive Aufbau filling: a filled d subshell ($3d^{10}$) plus a single $4s$ electron is favored. Thus longhand:

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$$

Check count: $2 + 2 + 6 + 2 + 6 + 10 + 1 = 29$.
Shorthand: core through 3p is argon, so [Ar]$3d^{10} 4s^1$. Note option (a) orders as $3d^{10} 4s^1$ but writing as [Ar]$4s^1 3d^{10}$ is also seen; the conventional shorthand groups the d after the noble gas core.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$; [Ar]$3d^{10} 4s^1$

Content (fully worked out problems & answers) continued...

(20) Write both the longhand and shorthand electron configurations for Mn^{2+}.

a. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; Shorthand: [Ar]$3d^5$

b. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$; Shorthand: [Ar]$4s^2 3d^3$

c. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$; Shorthand: [Ar]$3d^6$

d. Longhand: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^4$; Shorthand: [Ar]$4s^1 3d^4$

Solution:
Manganese, Mn, has $Z = 25$ (25 electrons) in the neutral atom. Mn^{2+} has lost 2 electrons $\Rightarrow$ electrons remaining $= 25 - 2 = 23$. Neutral Mn is commonly written [Ar]$4s^2 3d^5$. Remove the two $4s$ electrons first when forming cations from transition metals, leaving $4s^0 3d^5$. So the resulting configuration is:
Longhand: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$ Check: $2 + 2 + 6 + 2 + 6 + 5 = 23$.
Shorthand: [Ar]$3d^5$.

Answer: a. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$; [Ar]$3d^5$

Content (fully worked out problems & answers) continued...

(22) What is an electron orbital?

a. A circular path around the nucleus like planets orbiting the sun

b. A three-dimensional region where there is a high probability of finding an electron

c. A fixed energy value without any spatial meaning

d. A positively charged particle in the nucleus

Solution:
An orbital is not a fixed path (like planets), but rather a mathematical description of the region in space where an electron is most likely to be found. It is based on quantum mechanics and describes electron probability distributions.

Answer: b. A three-dimensional region where there is a high probability of finding an electron

Content (fully worked out problems & answers) continued...

(23) Which statement best describes the difference between an orbital and an orbit?

a. Both mean the same thing: a circular path around the nucleus

b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

c. An orbital is smaller than an orbit

d. An orbit describes the nucleus, while an orbital describes protons

Solution:
Bohr's model described electrons in fixed circular paths called orbits. Modern quantum mechanics replaces this idea with orbitals, which are probability regions where electrons are most likely found. Orbitals are not fixed paths, but probability clouds.

Answer: b. An orbit is a fixed circular path, while an orbital is a region of high probability for an electron

Content (fully worked out problems & answers) continued...

(24) Draw the electron diagram (orbital boxes) for neutral oxygen (O). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Oxygen has 8 electrons. Filling orbitals using Aufbau/Hund/Pauli:

2p	↑	↑	↑
2s	↑↓		
1s	↑↓		

Answer: $1s^2\ 2s^2\ 2p^4$; 2 unpaired electrons in 2p orbitals.

Content (fully worked out problems & answers) continued...

(25) Draw the electron diagram (orbital boxes) for the chloride ion, Cl^-. Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3p	☐	☐	☐
3s	☐		
2p	☐	☐	☐
2s	☐		
1s	☐		

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$. All orbitals fully filled:

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: Fully filled orbitals; closed shell.

Content (fully worked out problems & answers) continued...

(26) Draw the electron diagram (orbital boxes) for neutral iron (Fe). Place arrows (↑, ↓) to indicate electrons. Use Aufbau, Hund, and Pauli. Pay attention to how 4s and 3d are filled and later how electrons are paired/unpaired in 3d.

Empty orbital diagram (TikZ)

3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Solution: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$:

3d	↑ ↑ ↑ ↑ ↑↓
4s	↑↓
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Answer: 4 unpaired electrons in 3d.

Content (fully worked out problems & answers) continued...

(27) Draw the electron diagram (orbital boxes) for neutral copper (Cu). Place arrows (↑, ↓) to indicate electrons.

Empty orbital diagram (TikZ)

3d	☐ ☐ ☐ ☐ ☐
4s	☐
3p	☐ ☐ ☐
3s	☐
2p	☐ ☐ ☐
2s	☐
1s	☐

Solution: Cu has electron configuration [Ar] $3d^{10}$ $4s^{1}$:

3d	↑↓ ↑↓ ↑↓ ↑↓ ↑↓
4s	↑
3p	↑↓ ↑↓ ↑↓
3s	↑↓
2p	↑↓ ↑↓ ↑↓
2s	↑↓
1s	↑↓

Answer: $4s^{1}$ $3d^{10}$; 1 unpaired electron in 4s.

Content (fully worked out problems & answers) continued...

(28) Draw the electron diagram (orbital boxes) for the sulfide ion, S^{2-}. Place arrows (↑, ↓) to indicate electrons. Place arrows consistent with Aufbau, Hund, and Pauli.

Empty orbital diagram (TikZ)

3p	☐	☐	☐	
3s	☐			
2p	☐	☐	☐	Fill: 18 electrons; closed shell is expected.
2s	☐			
1s	☐			

Solution: Electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ (like Ar):

3p	↑↓	↑↓	↑↓
3s	↑↓		
2p	↑↓	↑↓	↑↓
2s	↑↓		
1s	↑↓		

Answer: All orbitals filled; closed-shell configuration.

Content (fully worked out problems & answers) continued...

(29) A tectonic plate moves 860 km per day. What is its speed in m/s?

a. 0.995 m/s

b. 9.95 m/s

c. 95.9 m/s

d. 0.0959 m/s

Answer:

$$\text{Speed} = \frac{860\ \cancel{\text{km}}}{\cancel{\text{day}}} \times \frac{1000\ \text{m}}{1\ \cancel{\text{km}}} \times \frac{1\ \cancel{\text{day}}}{86400\ \text{s}} = 9.95\ \text{m/s}$$

Content (fully worked out problems & answers) continued...

(30) Convert 750 mL to in^3. (Use 1 in = 2.54 cm)

a. 4.58 in^3

b. 45.8 in^3

c. 458 in^3

d. 0.458 in^3

Answer:

$$750 \text{ mL} = 750 \text{ cm}^3, \quad 1 \text{ in}^3 = (2.54 \text{ cm})^3 = 16.387 \text{ cm}^3$$

$$\text{Volume} = \frac{750 \text{ cm}^3}{16.387 \text{ cm}^3/\text{in}^3} = 45.8 \text{ in}^3$$

Content (fully worked out problems & answers) continued...

(31) Convert 88.0 kPa to atm. (Use 1 atm = 101.325 kPa)

a. 0.868 atm

b. 0.880 atm

c. 1.15 atm

d. 0.815 atm

Answer:

$$\text{Pressure} = \frac{88.0\ \cancel{\text{kPa}}}{1} \times \frac{1\ \text{atm}}{101.325\ \cancel{\text{kPa}}} = 0.868\ \text{atm}$$

Content (fully worked out problems & answers) continued...

(32) Convert 4.6 g/L to mg/dL.

a. 0.46 mg/dL

b. 4.6 mg/dL

c. 46 mg/dL

d. 460 mg/dL

Answer:

$$\frac{4.6\ \cancel{g}}{\cancel{L}} \times \frac{1000\ \text{mg}}{1\ \cancel{g}} \times \frac{1\ \text{dL}}{10\ \cancel{L}} = 46\ \text{mg/dL}$$

Content (fully worked out problems & answers) continued...

(33) A block has mass 245.0 g and dimensions 4.50 cm × 3.20 cm × 2.10 cm. Find its density in kg/m^3.

a. 8.10×10^2 kg/m^3

b. 8.10×10^3 kg/m^3

c. 8.10×10^4 kg/m^3

d. 8.10×10^5 kg/m^3

Answer:

$$V = (4.50)(3.20)(2.10)\ \text{cm}^3 = 30.2\ \text{cm}^3$$

$$\rho = \frac{245.0\ \text{g}}{30.2\ \text{cm}^3} = 8.10\ \text{g/mL} = 8.10\ \frac{\text{g}}{\text{cm}^3}$$

$$\text{Density} = \left(8.10\ \frac{\text{g}}{\text{cm}^3}\right) \times \frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{10^6\ \text{cm}^3}{1\ \text{m}^3} = 8.10 \times 10^3\ \text{kg/m}^3$$

Content (fully worked out problems answers) continued...

(33) A student weighs a sample of water and records 248.5 g. The actual mass is 250.0 g. What is the difference between the recorded mass and the true mass?

Solution:

$$\text{Difference} = \text{True mass} - \text{Recorded mass} = 250.0\ \text{g} - 248.5\ \text{g} = 1.5\ \text{g}$$

Answer: 1.5 g

Content (fully worked out problems & answers) continued...

(34) A 0.500 g sample of NaCl dissolves in 250.0 mL of water. What is the concentration in g/L?

a. 0.125 g/L

b. 2.00 g/L

c. 1.25 g/L

d. 0.250 g/L

Answer:

$$\text{Concentration} = \frac{0.500 \text{ g}}{0.250 \text{ L}} = 2.00 \text{ g/L}$$

Content (fully worked out problems & answers) continued...

(35) Convert 987.0 mbar to Pa.

a. 9.870 Pa

b. 9.87×10^2 Pa

c. 9.87×10^4 Pa

d. 9.87×10^5 Pa

Answer:

$$987.0 \text{ mbar} \times \frac{100 \text{ Pa}}{1 \text{ mbar}} = 9.870 \times 10^4 \text{ Pa}$$

Content (fully worked out problems & answers) continued...

(36) A sample occupies 250.0 mL and has a mass of 0.890 g. Calculate its density.

a. 3.56×10^{-3} g/mL

b. 0.890 g/mL

c. 3.56 g/mL

d. 35.6 g/mL

Answer:

$$\rho = \frac{0.890 \text{ g}}{250.0 \text{ mL}} = 3.56 \times 10^{-3} \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(37) A beaker contains 3.50 L of solution. Express this volume in cubic decimeters (dm^3) using unit factors.

a. 0.00350 dm^3

b. 3.50 dm^3

c. 35.0 dm^3

d. 350 dm^3

Answer:

$$3.50 \text{ L} \times \frac{10^{-3} \text{ dm}^3}{1 \text{ L}} = 0.00350 \text{ dm}^3$$

Content (fully worked out problems & answers) continued...

(38) A student measures 0.00250 kg of a substance. Express this in grams.

a. 0.00250 g

b. 0.0250 g

c. 2.50 g

d. 250 g

Answer:

$$0.00250 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} = 2.50 \text{ g}$$

Content (fully worked out problems & answers) continued...

(39) A 5.00 mL sample of water weighs 4.98 g. What is its density?

a. 0.996 g/mL

b. 1.00 g/mL

c. 1.02 g/mL

d. 1.10 g/mL

Answer:

$$\rho = \frac{4.98 \text{ g}}{5.00 \text{ mL}} = 0.996 \text{ g/mL}$$

Content (fully worked out problems & answers) continued...

(40) Which of the following is an isotope of carbon?

a. ^{12}C

b. ^{14}C

c. Both a and b

d. ^{16}O

Solution:
Isotopes are atoms of the same element with different mass numbers (different neutrons). Both ^{12}C and ^{14}C are carbon isotopes.

Answer: c. Both a and b

Content (fully worked out problems & answers) continued...

(41) Which subatomic particle determines the identity of an element?

a. Proton

b. Electron

c. Neutron

d. Nucleus

Solution:
The number of protons (atomic number) defines the element. Changing the number of protons changes the element.

Answer: a. Proton

Content (fully worked out problems & answers) continued...

(42) An atom of sodium (Na) loses one electron. What is the resulting ion?

a. Na^{-}

b. Na^{+}

c. Na^{2+}

d. Neutral Na

Solution:
Losing one electron creates a positive charge. Sodium forms Na^{+} when it loses one electron.

Answer: b. Na^{+}

Content (fully worked out problems & answers) continued...

(43) Which of the following are isotopes of the same element?

a. ^{1}H and ^{2}H

b. ^{12}C and ^{14}N

c. ^{16}O and ^{18}F

d. ^{23}Na and ^{24}Mg

Solution:
Isotopes are atoms with the same number of protons but different neutrons. ^{1}H and ^{2}H are hydrogen isotopes.

Answer: a. ^{1}H and ^{2}H

Content (fully worked out problems & answers) continued...

(44) Which bond is the most polar?

a. C–H

b. O–H

c. N–H

d. Cl–H

Solution:
Bond polarity depends on the difference in electronegativity between the two atoms. Approximate values: C (2.5), H (2.1), O (3.5), N (3.0), Cl (3.0).

$$\Delta EN(C\text{–}H) = 0.4, \quad \Delta EN(O\text{–}H) = 1.4, \quad \Delta EN(N\text{–}H) = 0.9, \quad \Delta EN(Cl\text{–}H) = 0.9$$

Thus, O–H has the largest electronegativity difference, making it the most polar bond.

Answer: b. O–H

Content (fully worked out problems & answers) continued...

(45) Which molecule is nonpolar overall despite having polar bonds?

a. CO_2

b. H_2O

c. NH_3

d. HCl

Solution:
A molecule is nonpolar overall if its bond dipoles cancel due to geometry. CO_2 has two polar C=O bonds but they are 180° apart in a linear structure, so the dipoles cancel. H_2O (bent), NH_3 (trigonal pyramidal), and HCl (diatomic) are all polar molecules.

Answer: a. CO_2

Content (fully worked out problems & answers) continued...

(46) Which type of bond is expected between Na and Cl in NaCl?

a. Nonpolar covalent

b. Polar covalent

c. Ionic

d. Metallic

Solution:
Electronegativity difference between Na (0.9) and Cl (3.0) is about 2.1, which is large. A difference greater than ~ 2.0 typically indicates ionic bonding. Thus, Na transfers an electron to Cl, forming an ionic bond.

Answer: c. Ionic

Content (fully worked out problems & answers) continued...

(47) Which of the following has the highest electronegativity?

a. Lithium

b. Carbon

c. Fluorine

d. Oxygen

Solution:
Electronegativity increases across a period and decreases down a group. Fluorine is the most electronegative element (value ≈ 4.0), higher than oxygen (3.5), carbon (2.5), and lithium (1.0).

Answer: c. Fluorine

Activity 1

Collaborative Learning Techniques	Assigned Discussion Leader
Learning Strategy	Family Feud
Time	12:10—12:30
Materials	Calculator, writing utensil, paper / tablet, conversion sheets, periodic table
Lecture/Textbook/Old Plans/Slide References	Chapter 2: Measurements, Chapter 3: Periodic Table

Definition

1. **Assigned Discussion Leader:** One person in the group is asked to present on a topic or review material for the group and then lead the discussion for the group. This person should not be the regular group leader.

2. **Family Feud:** This will work well as a test or quiz review for concepts and making connections between terms. Have students write down one term that comes to mind when a broad category is given. Create a list of the top 4 or 5 and use those as "answers" for the game. During the session, dividie studetns into two groups. Given the topic (broad category from the survey), students can decide wheher they want to play or pass. If they play, they alternate turns calling out a term they think is relevant. If they are correct, they get the points. If not, the next student attempts. Continue in this fashion until all terms are guessed and explained or three incorrect guesses are made. If three incorrect guesses occur, the other team tgets the chance to finish the topic.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are organized into teams and reminded that the game covers unit conversions, electron configurations, isotopes, elements, and the periodic table.
2. Each team takes turns selecting a category and point value. The team has a set time (e.g., 30–60 seconds) to discuss and provide an answer.
3. The SI leader monitors discussions, ensuring all students participate and encouraging teams to explain their reasoning before answering.
4. If a team answers incorrectly, other teams have the opportunity to discuss and "steal" the question.
5. Points are awarded for correct answers and tracked on the board or a visible scoreboard, helping students see progress and encouraging friendly competition.
6. Between rounds, the SI leader pauses to clarify any misconceptions, review key concepts, and provide real-life examples to connect the question to practical applications.
7. After all questions are answered, the SI leader leads a short debrief, asking teams to reflect on what concepts were challenging, what strategies helped, and how preparation and teamwork influenced performance.

Checking for Understanding (general conceptual questions)

1. How do you approach converting between different units of length, volume, or pressure?
2. What strategies can you use to convert between metric and imperial units?
3. How do you determine the density of a substance, and why is knowing density useful?
4. What patterns or trends exist in the periodic table that help classify elements?
5. How can you tell whether an element is a metal, nonmetal, or metalloid?
6. What defines a group or a period on the periodic table?
7. How does the number of protons, neutrons, and electrons affect the identity and properties of an atom?
8. What is the difference between an atom and an ion, and how do ions form?
9. How do isotopes of the same element differ, and what stays the same?
10. How do electron configurations help predict chemical behavior of elements?
11. What is the difference between an orbital and an orbit, and why is this important for understanding electron arrangements?
12. How do you determine if a chemical bond is ionic, polar covalent, or nonpolar covalent?
13. How can you tell if a molecule is polar or nonpolar overall?
14. Why are electronegativity differences important in predicting bond polarity?
15. How do concentrations relate to the mass of solute and volume of solution?
16. How do scientific notation and powers of ten help when dealing with very large or very small quantities?

Activity 2

Collaborative Learning Techniques	Fishbowl (many), Jigsaw (few)
Learning Strategy	Boardwork Model (modified)
Time	12:30—12:50
Materials	Colored whiteboard markers, a whiteboard, partitioned questions (to display multiple to multiple groups at a time).
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Fishbowl:** The group is separated into an "inside" and an "outside" the fishbowl. Students inside the fishbowl will participate in the activity or discussion. Students outside the fishbowl will listen to the discussion and observe the students engage in the activity. After the fishbowl, students inside are given the opportunity to reflect on what they observed, provide feedback to those isnide, and can contribute more to the discussion.

2. **Jigsaw:** Jigsaws, when used properly, make the group as a whole dependent upon all of them in subgroups. Each group provides a peice of the puzzle. Group members are broken into smaller groups. Each small group works on some aspect of the same problem, question, or issue. They then share their part of the puzzle with the large group.

3. **Boardwork Model:** This is a method of organizing boardwork in order to fascilitate an undertsanding of problem-solving strategies. The baord should be divdied into 4 sections: 1.) Prerequisite Knowledge, 2.) Mathematical Steps, 3.) Narrative of the Steps, 4.) Additional Sample Problem. Encourage one student to fill out section 1 on the board. Then encourage 2 students to simultaneously complete section 2 and 3 on the board. Lastly, have another student complete section 4. Encourage the students to use this model when studying outside of the SI session.

Instructions (please provide aDETAILED activity breakdown below (and/or attached)

1. **Objective:** Guide students to work in groups of at least 2 to solve chemistry-based problems collaboratively on a whiteboard.

Setup

- Arrange the classroom so multiple groups can work at the whiteboard simultaneously.
- Provide each group with markers and a section of the whiteboard.
- Prepare a set of chemistry problems in advance.
- Optionally, provide reference tables or hints for problem-solving to keep groups on track.

Running the Activity

1. Introduce the activity and explain that groups of at least 2 will calculate problem results together.
2. Present the first chemistry problem to all groups.
3. Groups collaborate to solve the problem on the whiteboard.
4. Rotate through problems, allowing each group to contribute their calculations and drawings.
5. Monitor group progress, providing hints or clarifications as needed without giving away solutions.
6. Encourage discussion and peer teaching within groups to reinforce chemistry concepts.
7. Continue until all problems are solved and the final picture is completed.

Facilitator Role

- Ensure groups are on task and collaborating effectively.
- Clarify instructions and answer conceptual questions without solving problems for the students.
- Track timing to ensure the activity progresses smoothly.
- Circulate to observe problem-solving strategies, noting areas where students struggle for future review.
- Celebrate the completion of the final picture and debrief with the class on the chemistry concepts applied.

Checking for Understanding (questions & answers)

Questions:

1. Why does the boardwork model provide a calmer, low-pressure environment for students?
2. How does the centralized canvas in the boardwork model help reduce student anxiety about mistakes?
3. In what ways can students approach chemistry problems when using the boardwork model?
4. What types of chemistry tasks can students work through step by step with this model?
5. How do peers act as resources in the boardwork model?
6. How does peer explanation and feedback reinforce understanding of chemistry concepts?
7. Within a Supplemental Instruction (SI) context, how does the boardwork model encourage participation?
8. What benefits do students gain from multiple perspectives on the same chemistry problem?
9. How does the shared board help students practice applying theoretical knowledge in a visual way?
10. What role does the SI leader play when using the boardwork model during a session?
11. How does this model help the SI leader identify common misconceptions?
12. In what way does the boardwork model make SI sessions more effective for learners of varied skill levels?

Closer

Collaborative Learning Techniques	Group Survey
Learning Strategy	Brain dump
Time	12:50—12:55
Materials	Index cards, writing utensils
Lecture/Textbook/Old Plans/Slide References	Chapter 1: Basics of Matter, Chapter 2: Measurements

Definitions

1. **Group survey:** Each group member is surveyed to discover their position on an issue, problem or topic. This process ensures that each member of the group is allowed to offer or state their point of view.

2. **Brain dump:** This is a great closer for students to sum up their knowledge on the topic(s) from the session. Have the students take out a blank piece of paper /Google Doc and write/type("dump") everything they know relevant from the discussion that day. The SI Leader can look at their work to assess the students' understanding of the course material and have them compare it to each other. It is important to at least discuss with the students so the SI Leader knows the students' comfort levels with the material.

Instructions (please provide aDETAILED activity breakdown below (and/or attached

1. Students are put into small groups and are asked to write down their confidence on a single notecard without names.
2. The cards are evidence of the group's confidence in the content.

Content (fully worked out problems & answers)

- No new worked problems; brain dump.

Checking for Understanding (questions & answers) continued...

- Review confidence levels
- Interpret the variety of confidence and their distribution in groups.

Plan Checklist

- ☐ Variety of Collaborative Learning Techniques and Learning Strategies
- ☐ Chapter/slide# where information is coming from
- ☐ Included timestamps (and buffer time as needed)
- ☐ Details on how leader will break up students
- ☐ Checking for Understanding questions
- ☐ Fully worked out problems and examples for each activity
- ☐ Necessary links required for online implementation (if applicable)
- ☐ Source material correctly cited (if applicable)
- ☐ Plan is uploaded to Teams